

Energy-intensive Industries - Circularity and Zero Pollution

Proposals are invited against the following topic(s):

HORIZON-CL4-2025-05-TWIN-TRANSITION-35-two-stage: Developing and embedding upcycling technologies into viable business (Processes4Planet partnership) (IA)

Call: INDUSTRY two-stage	
Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of between EUR 8.00 and 12.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 48.00 million.
<i>Type of Action</i>	Innovation Actions
<i>Admissibility conditions</i>	The conditions are described in General Annex A. The following exceptions apply: Applicants submitting a proposal under the blind evaluation pilot (see General Annex F) must not disclose their organisation names, acronyms, logos nor names of personnel in the proposal abstract and Part B of their first-stage application (see General Annex E).
<i>Technology Readiness Level</i>	Activities are expected to achieve TRL 7 by the end of the project – see General Annex B.
<i>Procedure</i>	The procedure is described in General Annex F. The following exceptions apply: This topic is part of the blind evaluation pilot under which first stage proposals will be evaluated blindly. To ensure a portfolio including support leading to the recovery and upgrading of industrial infrastructure in Ukraine, grants will be awarded to applications not only in order of ranking, but also to at least one project involving at least one non-profit partner and at least one industrial partner established and located in Ukraine, provided that the application attains all thresholds.
<i>Legal and financial set-up of the Grant Agreements</i>	The rules are described in General Annex G. The following exceptions apply:

	Eligible costs will take the form of a lump sum as defined in the Decision of 7 July 2021 authorising the use of lump sum contributions under the Horizon Europe Programme – the Framework Programme for Research and Innovation (2021-2027) – and in actions under the Research and Training Programme of the European Atomic Energy Community (2021-2025) ³¹ .
<i>Exceptional page limits to proposals/applications</i>	In order to include a business case and exploitation strategy, as outlined in the introduction to this Destination, the page limit in part B of the General Annexes is exceptionally extended by 3 pages.

Expected Outcome: Energy Intensive industries will benefit from the following outcomes:

- Prove the technical, economic, and environmental feasibility of the upcycling of end-of-life waste materials by process industries in integrated circular schemes – including for supplying the value chains of net zero technologies and components;
- Enable doubling the ratio of secondary raw materials upcycled leading to a significant increase in resource, including energy efficiency and improved carbon lifecycle across the value chain compared to present levels;
- Increase the competitiveness of the European process industry, including by providing new business opportunities and revenue flows for recycling companies benefiting particularly SMEs – including for supplying the value chains of net zero technologies and components;
- Foster the use of digital tools as well as the data sharing, and FAIR (Findability, Accessibility, Interoperability and Reusability) digital assets principles.
- As appropriate in one or more projects, contribute to the reconstruction, recovery, circularity and upgrading of industries of Ukraine.

Scope: The recycling of end-of-life materials into valuable materials that can replace primary raw materials requires integrated systems from the collection to dismantling and separation to their final processing. The technologies and implementation (including logistics and economics) of such circular schemes should be addressed in an integrated way. The innovation needed will depend on the addressed waste category. However, even if the upcycling technologies and implementation may be sector/material specific, the cross-sectorial elements are important and should deserve due attention.

Proposals are expected to address all of the following:

³¹ This [decision](https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf) is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under ‘Simplified costs decisions’ or through this link: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf